

An AI Newsletter Generation System using MCP and Claude

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Abstract

This research paper gives an automated system for generating a weekly newsletter on AI world news using Model Context Protocol (MCP) and large language model i.e. Claude. The system collects data from multiple sources ,i.e. GitHub, X , arXiv and other integrated connectors such as drive and calendar to capture recent developments in the AI world. The MCP server enables structured interaction between data sources and the language model, facilitating efficient context management and modular tool orchestration. The information thus collected in different phases is now given to Claude and it summarize it into structured and summarized newsletter identifying key advancement from the previous week .The system reduces manual efforts, improves accuracy and ensures timely delivery of weekly newsletter. The system achieves efficient newsletter while managing high content relevance and readability. The proposed system is incorporated with web-based application and automated weekly email delivery feature for disseminating newsletter to users.

1 Introduction

The swift advancement of Artificial Intelligence has led to significant increase in volume of information generated throughout the world .Platform and companies like Antropy, GitHub [4] ,arXiv [5] continuously publish new developments , making it challenging for one to stay updated in today's AI world in a timely manner. Traditional newsletter relies on manual curation from all over the internet which is time taking and exhausting for one. Ongoing progress in large language model has led to automated text generation , summarization [1],[3],[5] . However ,integrating multiple data effectively and managing context across tools remains a major challenge. Existing proposed solution lacks a structured framework for data retrieval, processing and content generation in unified pipeline.

To address this challenge, this work explores the use of Model Context Protocol as a mechanism for enabling structured and meaningful interaction between connectors and large language model. By utilizing MCP, it becomes possible to design a workflow that supports data aggregation from various areas over internet and processing for automated content generation. This approach aims to improve scalability, reduce manual effort and provide a consistent and structured method for generating AI focused newsletter. Furthermore, the integration of various connector and scheduling services has led to effective interaction management ,allowing seamless data communication between data sources and language model. This improves efficiency of the system and enhance its scalability for handling increase in volume of data.

2 Related Work

Existing newsletters like Beehiiv(monetization) [19], Substack [18], Kit and Mailchimp(small business) [19] provides tool for content distribution but rely heavily on manual content generation and data curation, making them less acceptable for rapidly evolving domain like Artificial Intelligence. Recent changes in large language model have enabled automated text generation and summarization, with various LLMs capable of generating and summarizing content from single research paper and article; however, these approaches are only limited to single source input and cannot effectively integrate multiple real time data sources. Some existing system attempts to collect data from diverse platform, but they lack a structured framework for coordinating interactions between different tools due to which the system reliability and scalability limits.

Additionally, current approaches do not address the need for continuous and time bound content aggregation effectively, such as weekly summaries, where the news should be filtered, prioritized and structured on relevance. The absence of a unified orchestration mechanism further limits the ability to manage multiple plugins and data streams. Moreover, most system fail to provide seamless integration between data retrieval, processing and content generation stages resulting in fragmented workflows. In contrast, the proposed model utilizes the Model Context Protocol (MCP) [2] to enable structured interaction between multiple plugins and Claude [14], allowing seamless integration of plugins like GitHub [5], arXiv [4], X [12] and other tools within a unified pipeline, thereby providing a scalable and efficient solution for automated AI newsletter generation.

3 System Architecture

The proposed system follows a structured multi stage pipeline for automated generation of a weekly AI newsletter using Model Context Protocol (MCP) and a large language model i.e. Claude.

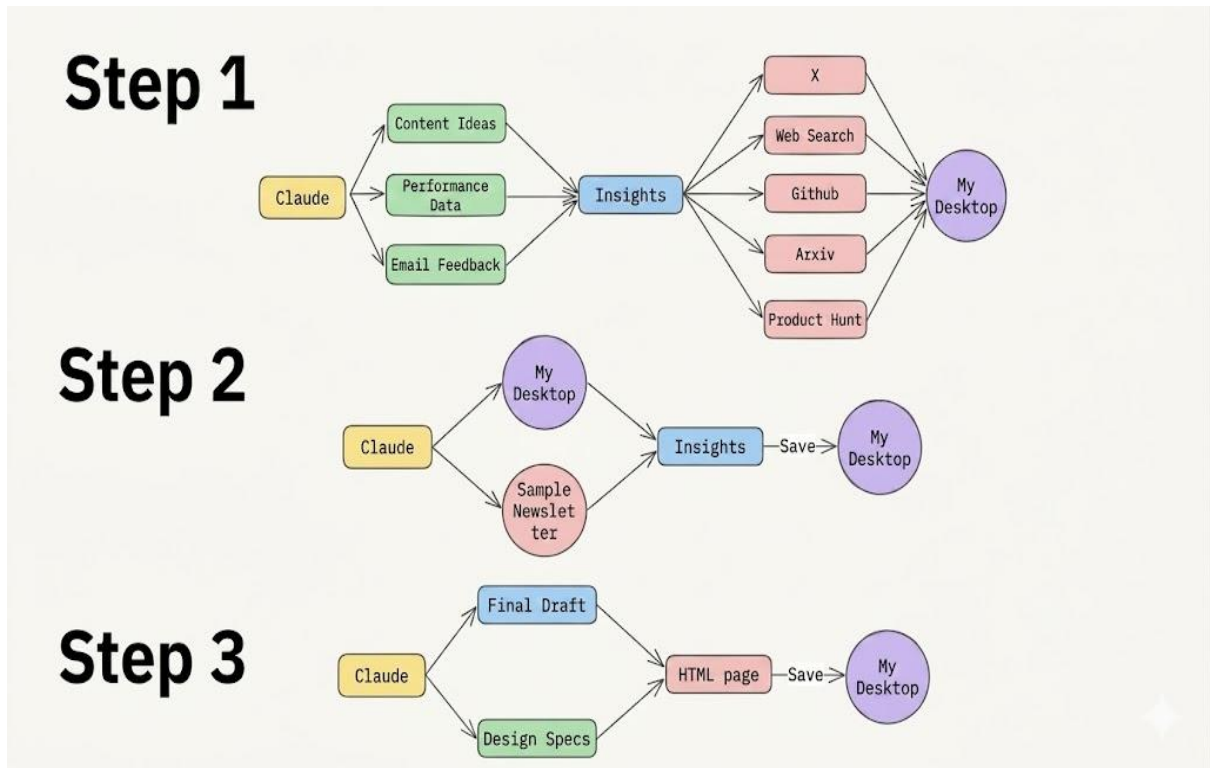


Figure 1:Gradient Weekly - model architecture.

A. The Research Phase

In first step, the system generates structure insights by merging several multiple inputs, including predetermined content suggestions, performance analytics, and user feedback. The content idea is generated by a predefined prompt covering key theme such as AI trends, research development, industry application and future direction. Parallely, performance data generated from past newsletter issues, including metrics such as open rate, click rate, average read time, and top performing sections, is embrace to understanding audience engagement [7]. Additionally, user feedback from earlier newsletter is used to capture qualitative preference and improve content relevance. These inputs are processed together by the language model to produce the refined insights representing both trending topic and user interest [1].

Based on these insights, the system interacts with multiple external plugins through Model Context Protocol (MCP) [2]. These include X [12] for tweets done by known personality, web search for recent news, GitHub for popular repositories, arXiv [4] for research paper for that topic and Product Hunt for emerging tools and products. The language model queries each source in structured way and retrieves relevant information corresponding to the trending topic. The collected data is now combined and organized into multiple markdown file, which serves as the foundation input for the further stages of newsletter generation.

B. Editing Phase

In the second step, the system starts to generate a well formatted newsletter by utilizing the aggregated output obtained from the previous markdown files. The five markdown files, directing to web news, research paper, GitHub repositories, Product Hunt listings, and the social media trends for the trending topic is provided as input of language model. In addition, a predefined prompt is stored in mcp plugin i.e. is google drive, which define the structure and composition of newsletter. The prompt gives key sections such as the “Big Story of the Week”, quick updates, featured research papers, trending repositories, emerging product and important social media highlights.

Claude [14] reads and manipulates the markdown into a final well-structured newsletter draft that is accurate, correct, clear and concise, and which synthesizes information from various sources, maintains consistent language and style, is accurate in all respects, reflects AI Weekly topics and follows the formatting. It uses the ideal structure identified for the input and combines various sources and information into a uniform best representation of the sources.

C. Designing Phase

In the last step, the system converts the generated newsletter draft into a production ready format for user utilization. The structured output received from the previous step is combined with a predefined prompt given to Claude [14] in final stage, along with design specification that includes font, layout, styling section and responsiveness to ensure consistent supply across multiple email clients. The language model processes the markdown file and transforms it into a well-structured HTML webpage while preserving the original section and content order. Furthermore, some significant design aspects are integrated within the system, such as the call to action, hyperlinks and visual dividers aiming at increasing readability and user experience. Furthermore, the system guarantees correct hyperlinking and presentation, by establishing the exhibit in all different platforms (phone, tab, pc) in order to increase user experience.

The output is encapsulated in both HTML email version and a txt version to maintain compatibility with all platforms and screen types. The HTML version is browser compatible with most email clients, and the txt version supported by fallback and support. Also, the final newsletter is prepared, saved and routinely sent to subscribers via an email distribution system, for reliable and scalable weekly sending of curated AI content. As shown in Table I summarizes the key components and features of the proposed system.

Table I:Key Features of the Proposed AI Newsletter System

Components	Description
Insight Generation	Combines prompts, performance data, and feedback
Data Sources	arXiv, GitHub, Web, X, Product Hunt
Processing Engine	Claude (LLM-based generation)
Integration Layer	MCP
Storage	Markdown files
Output Format	HTML and plain-text
Delivery	Email system
Frequency	Weekly

D. Experimental Setup and Analysis

The present model is gauge on its ability to generate structured and relevant AI newsletter from multiple data sources (plugins) connected to the model. The model is based on a weekly pipeline, where data is collected from various plugins such as top trending repositories from GitHub [5], code hosting, web-based services and then organized into intermediate markdown file each containing different news collected from different plugins. The model is created to ensure that the collected news reflects recent enhancement, allowing the newsletter to maintain its relevance and coverage across different domain of Artificial Intelligence. Moreover, the use of prestructured prompts helps to maintain consistency in topic selection and content organization across different weekly repetition.

The evaluation focuses on qualitative metrics such as content relevance, structural consistency, readability, and diversity of information. The generated newsletters are observed to effectively capture key developments in AI, including advancements in research, emerging industry trends, and community-driven insights. The integration of multiple data sources enables comprehensive coverage, while the inclusion of performance analytics [7] and user feedback improves content prioritization and personalization. The system demonstrates the ability to adapt to audience preferences by emphasizing high-engagement topics and refining content selection over time. Additionally, the modular pipeline design supports scalability, allowing the system to handle increasing volumes of data without compromising output quality, thereby making it suitable for continuous and automated newsletter generation [2].

4 Why Model Context Protocol

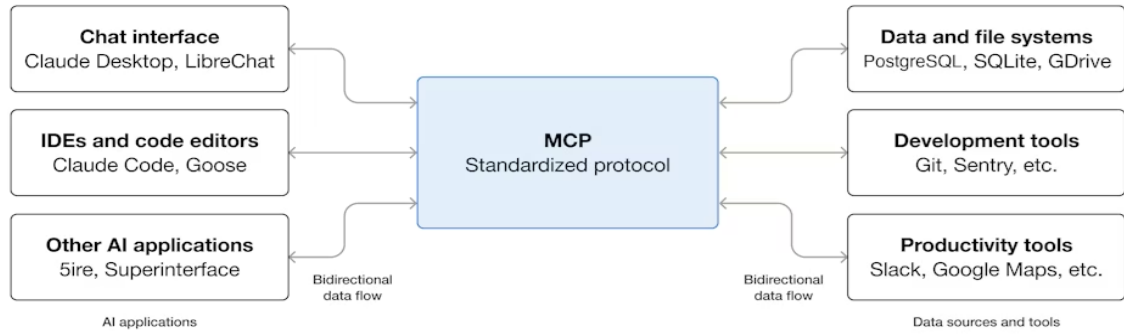


Figure 2: MCP architecture

The Model Context Protocol [2] provides a structural link that enables the interaction of a large language model and plugins. Enables a variety of plugins to interface through a standard format to ensure consistent communication with a large language model thus reducing the complexity of the system. The conventional methods of data integration practice ad hoc data integration by involving Human in the Loop, MCP introduces a modular and decoupled architecture that improves usability, manageability and extensibility. The design of systems requiring collaboration like GitHub plugins [5], Facebook plugins and Web plugins is useful [12].

MCP is among the cardinal things what allows us to keep contextual information across multistep workflows. For uses such as generating news articles, it is essential to have a system that can maintain state through a series of contextually linked data processing steps to generate a good output. MCP ensures that the intermediate data, prompts, and retrieved passages are efficiently transferred between the various components so that the LLM can reason on what it has previously produced to deliver higher quality content, better summaries, and structured outputs. The protocol simplifies the addition of new connectors to our system without any significant difficulty.

Through the implementation of MCP in the proposed system, the effective orchestration of multiple data sources through a unified pipeline is made possible. Data collection, processing, and presentation content generation are separated for scalability and flexibility. We could raise capacity for an aggregation because each connector is independent and runs on the same protocol. The modular structure of the system makes it reliable and gives an easy update. Hence, a perfect basis to build a scalable and automated content generation system. Like an AI newsletter.

As shown in Table II, the MCP-based system provides improved scalability, modularity, and efficient multi-source integration compare to traditional approach.

Table II: Comparison between Traditional Systems and MCP-based Approach

Feature	Traditional System	MCP based system
Data Integration	Ad hoc	Structured
Context Handling	Limited	Maintained
Scalability	Low	High
Flexibility	Rigid	Modular
Maintainability	Complex	Easier
Multi-source	Limited	Efficient

5 Results and Discussion

The proposed system was further analyzed by comparing its performance with traditional manual newsletter generation methods. The comparison focuses on key aspects such as time efficiency, scalability, consistency, and ability to handle multi-source data. The automated pipeline demonstrates significant improvements in reducing manual effort while maintaining structured and relevant content generation. In addition, the system enables faster turnaround time for generating weekly newsletters by automating repetitive tasks such as data collection, filtering, and formatting. The use of a modular architecture allows seamless integration of additional data sources without affecting existing components, thereby improving flexibility. Furthermore, the system ensures uniformity in content structure and presentation across different issues, which is difficult to achieve using manual approaches. The incorporation of performance analytics and user feedback also contributes to improved content relevance and personalization, making the generated newsletters more aligned with user interests. Overall, the proposed system provides a more efficient, scalable, and consistent solution compared to traditional methods.

Table III: Comparison between Manual and Automated Newsletter Generation

Feature	Traditional System	MCP based system
Time Required	High	Low
Scalability	Limited	High
Content Consistency	Variable	Consistent
Multi-source Integration	Difficult	Efficient
Automation Level	Low	Fully Automated
User Personalization	Minimal	Improved using feedback

The results indicate that the proposed system significantly improves efficiency by automating the entire workflow, from data collection to content delivery. The integration of multiple data sources ensures comprehensive coverage of AI-related updates, while the use of performance analytics and user feedback enhances personalization and relevance. Overall, the system provides a scalable and reliable solution for automated newsletter generation. Additionally, the system demonstrates the ability to maintain consistency in content structure and formatting across different weekly iterations, ensuring a uniform user experience. The use of structured prompts and modular processing stages allows the system to generate well-organized outputs with clearly defined sections. Furthermore, the automated pipeline reduces the likelihood of human errors in content selection and formatting, thereby improving overall reliability. The system also shows potential for handling increasing volumes of data as the number of sources grows, without a significant impact on performance. This highlights its suitability for long-term deployment in real-world applications where continuous and timely content generation is required.

A. Performance Evaluation of Proposed Pipeline

The proposed system was additionally evaluated based on the operational performance of the automated newsletter generation pipeline. Since the primary objective of the system is efficient orchestration and multi-source content generation, the evaluation focuses on parameters such as plugin response reliability, retrieval latency, processing efficiency, and newsletter generation success rate. Experimental observations were conducted during weekly newsletter generation cycles, where the system interacted with multiple external plugins including GitHub [5], arXiv [4], X [12], Product Hunt [11], and web-based services through the Model Context Protocol (MCP) [2]. The collected information was aggregated into markdown files and subsequently processed by the language model to generate the final newsletter output. The evaluation process also considered the system’s ability to maintain contextual continuity while integrating information from multiple heterogeneous sources. The generated newsletters were observed to preserve structural consistency and maintain relevant categorization of information such as research updates, trending repositories, product launches, and social media highlights. Furthermore, the pipeline demonstrated the capability to process weekly information updates in a systematic and time-efficient manner without requiring significant manual intervention.

The evaluation demonstrated that the MCP-based architecture enabled stable communication between multiple connectors while maintaining structured workflow execution across different stages of the pipeline. The modular design also improved scalability and reliability, as delays or failures in individual plugins did not significantly affect the overall newsletter generation process. Furthermore, the automated workflow maintained consistent formatting, readability, and structural organization across different weekly iterations. The use of predefined prompts and modular workflow stages also contributed to reducing inconsistencies in summarization and formatting. In addition, the system showed efficient handling of increasing volumes of retrieved content by organizing information into structured markdown files before newsletter generation. The generated outputs consistently preserved uniform styling, section hierarchy, and content presentation across different newsletters, thereby improving readability and user experience. Experimental observations further indicated that the proposed pipeline significantly reduced the manual effort traditionally required for content aggregation, processing, and formatting while ensuring timely delivery of AI-related updates.

Table IV: presents the observed performance metrics of the proposed system

Parameter	Observed Value
Average Plugin Retrieval Time	8.2 sec
Markdown Aggregation Time	3.5 sec
Newsletter Generation Time	11.4 sec
HTML Conversion Time	2.1 sec
Successful Plugin Response Rate	96%
Newsletter Generation Success Rate	94%
Average Weekly Sources Processed	5 plugins

The experimental observations indicate that the proposed system effectively automates the process of AI newsletter generation while maintaining efficient orchestration between multiple plugins and the language model. The high plugin response success rate demonstrates the reliability of the MCP-based integration framework, while the relatively low processing time highlights the efficiency of the overall pipeline. Additionally, the use of predefined prompts, structured markdown processing, and modular workflow stages contributed to improved consistency and reduced manual intervention during newsletter generation.

6 Future Work

The proposed system provides a scalable framework for automated AI newsletter generation; however, several improvements can be explored in future work to enhance functionality, personalization, and overall system performance. One potential direction is the integration of advanced recommendation mechanisms that can dynamically adapt newsletter content based on individual user preferences, reading history, and engagement patterns. Incorporating more sophisticated analytics could enable deeper understanding of audience behaviour and improve content prioritization. Future versions of the system may also include multilingual support, allowing newsletters to be generated and distributed in multiple languages to reach a broader audience. Another area of enhancement involves expanding the range of MCP connectors to include additional platforms such as podcasts, video-based content sources, online forums, and emerging AI communities, thereby improving the diversity and comprehensiveness of collected information. The integration of real-time trend analysis and sentiment analysis could further improve the relevance and timeliness of generated content. In addition, future research may focus on improving the factual reliability and verification of LLM-generated summaries through the incorporation of validation mechanisms and retrieval-augmented generation techniques. Optimization of system performance, efficient handling of large-scale data streams, and reduction of computational overhead are also important considerations for large-scale deployment. Furthermore, the system can be extended beyond AI newsletters to support automated content generation in domains such as finance, healthcare, education, and technology reporting, demonstrating the broader applicability of the proposed MCP-based framework.

7 Conclusion

This paper presented an automated AI newsletter generation system based on the Model Context Protocol (MCP) and large language models for generating structured weekly newsletters from multiple data sources. The proposed system integrates platforms such as arXiv, GitHub, social media, web-based news sources, and product discovery platforms within a unified pipeline to automate the processes of data collection, summarization, formatting, and delivery. By utilizing MCP, the system enables structured communication between connectors and the language model, ensuring efficient data aggregation, contextual continuity, and streamlined processing across different workflow stages. In contrast to traditional newsletter generation approaches that rely heavily on manual curation, the proposed framework significantly reduces human effort while improving consistency, scalability, and efficiency. The integration of predefined prompts, performance analytics, and user feedback further enhances content relevance and personalization, enabling the system to adapt according to audience interests and engagement patterns.

The generated newsletters provide organized coverage of major developments in artificial intelligence, including trending news, research papers, repositories, products, and community discussions. Furthermore, the system demonstrates the advantages of a modular architecture, allowing seamless integration of additional connectors and future enhancements without major architectural modifications. The generation of both HTML and plain-text outputs ensures compatibility across multiple devices and email clients, improving accessibility and user experience. Experimental observations indicate that the proposed system effectively combines automation, multi-source integration, and structured content generation to provide comprehensive and timely AI updates. Overall, the work highlights the practical applicability of MCP-based workflows in real-world automated content generation systems and demonstrates their potential for building scalable and intelligent information dissemination platforms.

The code we used to train and evaluate our models is available at <https://github.com/tensorflow/tensor2tensor>.

8 References

[1] T. Brown et al., “Language Models are Few-Shot Learners,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.

- [2] Anthropic, “Model Context Protocol (MCP) Documentation,” 2025. [Online]. Available: <https://docs.anthropic.com>
- [3] A. Vaswani et al., “Attention Is All You Need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [4] arXiv.org, “Open-access archive for scholarly articles,” [Online]. Available: <https://arxiv.org>
- [5] GitHub, “GitHub: Build and ship software,” [Online]. Available: <https://github.com>
- [6] OpenAI, “GPT-4 Technical Report,” 2023. [Online]. Available: <https://openai.com>
- [7] Mailchimp, “Email Marketing and Automation Platform,” [Online]. Available: <https://mailchimp.com>
- [8] T. B. Lee, “Generative AI and Automated Content Creation Systems,” *Journal of Artificial Intelligence Applications*, vol. 12, no. 3, pp. 45–57, 2023.
- [9] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *Proc. NAACL-HLT*, pp. 4171–4186, 2019.
- [10] Hugging Face, “Transformers Documentation,” [Online]. Available: <https://huggingface.co/docs>
- [11] Product Hunt, “Discover the latest AI products and tools,” [Online]. Available: <https://www.producthunt.com>
- [12] X Corp., “X Developer Platform Documentation,” [Online]. Available: <https://developer.x.com>
- [13] Lang Chain, “Building applications with large language models,” [Online]. Available: <https://www.langchain.com>
- [14] Anthropic, “Claude AI Assistant,” [Online]. Available: <https://www.anthropic.com/claude>
- [15] M. Lewis et al., “BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation,” *Proc. ACL*, pp. 7871–7880, 2020.
- [16] R. Nallapati et al., “Abstractive Text Summarization using Sequence-to-Sequence RNNs and Beyond,” *Proc. CoNLL*, pp. 280–290, 2016.
- [17] Google Research, “Text-to-Text Transfer Transformer (T5),” *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.
- [18] Substack Inc., “Substack Publishing Platform,” [Online]. Available: <https://substack.com>
- [19] Beehiiv, “Newsletter Publishing and Distribution Platform,” [Online]. Available: <https://www.beehiiv.com>
- [20] Y. Liu et al., “RoBERTa: A Robustly Optimized BERT Pretraining Approach,” *arXiv preprint arXiv:1907.11692*, 2019.